QUICK INSIGHTS

Artificial Intelligence Weather Prediction Models

RESEARCH QUESTIONS

What is an artificial intelligence weather prediction (AIWP) model and how do such models compare to traditional numerical weather prediction (NWP) models? What are the potential impacts of AIWP models on the operations of the power system and how should system operators be preparing?

KEY POINTS

- The power system is significantly impacted by the weather and this dependency is growing. The primary reasons for this are increases in electrification and wind and solar generation. Weather forecasts and other weather model data are thus a crucial component of system planning and operations.
- In the past most weather forecasts and modeled historical weather data have been produced by NWP models. These physics-based models require enormous computational resources to run quickly enough.
- AIWP models are completely data driven and use deep learning architecture to learn patterns from historical weather data to produce forecasts.
- Pretrained AIWP models can be 1000–10,000 times faster than conventional NWP models.
- Conversely, the training of AI weather models requires tremendous amounts of computation power and large amounts of training data. Training data are typically obtained from NWP outputs. Retraining is challenging.
- Unlike NWP models, pretrained AIWP models have canned configurations and cannot easily be customized to particular use cases. Input and output variables, resolution, and output interval are all predefined.
- Pretrained models and source codes are open sourced and available on GitHub. The pretrained models can be run on central processing units (CPUs) and graphics processing units (GPUs) but are significantly faster when running on GPUs.
- AIWP model output will become an important input to power system tools that use weather data to improve forecast accuracy and timeliness, and, through large ensembles, will provide more insight into uncertainty.

INTRODUCTION – WHAT ARE AI WEATHER MODELS?

The power system is, and always has been, significantly impacted by the weather. However, this dependency is rapidly growing in extent and complexity. While many factors are driving this, two primary ones are the increasing shares of wind and solar generation, and electrification of space conditioning and transportation. Thus, weather forecasts are an important component of system operations and are becoming increasingly critical as they are the main inputs into load and renewable generation forecasts. In addition, more weather data are now needed for planning; such data must usually be synthesized from weather models.

Historically, operational weather forecasts have been based on numerical weather prediction (NWP). For more information about NWP, see the EPRI report *Guidance on Numerical Weather Prediction: Models and Data Sources*.[1] However, around 2020 a new technology, AIWP, began to emerge as a potential alternative to NWP. In a few short years, it has advanced rapidly enough to now rival traditional NWP for some forecasting applications, and it is important to understand how this technology may disrupt weather forecasting in the utility space.

ARTIFICIAL INTELLIGENCE WEATHER PREDICTION VS. NUMERICAL WEATHER PREDICTION VS. MACHINE LEARNING FORECASTS

Both AIWP and NWP weather forecast models are prognostic models that take an estimate of the initial state of the atmosphere, predict how that state will evolve in time, and then use the future state as the new initial state to iterate the predictive process forward in time. When this process is done fast enough, it is possible for the model to predict future states before they have happened, forming the basis for a weather forecast. The difference between AIWP and NWP is in the way the future state is determined. NWP models apply the laws of physics that govern atmospheric motions and processes to estimate the future state mathematically. AIWP models, on the other hand, are data driven and use deep learning architectures such as graphical neural networks. Using these sophisticated models, together with huge datasets of consistently produced historical weather patterns that relate different points in time, AIWP can "learn" the complex relationships between each variable in time and space. Such a process essentially embeds the physics governing how the atmosphere behaves into the neural network in the same way one learns by trial and error how Newtonian mechanics impacts where a tossed ball will land and how high it will bounce.

The accuracy of both methodologies is dependent on how accurately the initial state of the atmosphere is represented – both in terms of how accurate and completely it is measured and in terms of the ability of the model to represent this state. However, the approach used to determine the future state is profoundly different. An AI model must be trained with sufficient data that it can intuit how different patterns evolve and how variables are connected, which means that the output of the model is only as good as the quantity and quality of the data it is trained on. Meanwhile, the accuracy of the NWP model is dictated by how well the physical laws can be represented and numerically solved. The availability of past data to learn from is irrelevant, other than to help model developers validate and tune the system and to provide a means to perform post processing.

AIWP is distinct from the types of machine learning used to produce load and renewable energy forecasts or to bias correct or otherwise post process NWP forecasts. While both employ machine learning methods, AIWP seeks to provide an alternative to NWP forecasting methods and is a relatively new development enabled by advances in the processing power needed for model training. Today's load and renewables forecasts typically use machine learning methods such as artificial neural networks, decision trees,

ACRONYMS	
AI	Artificial Intelligence
AIWP	Artificial Intelligence Weather Prediction (model)
CONUS	Continental United States
ECMWF	European Centre for Medium-Range Weather Forecasting
ERA5	ECMWF Reanalysis v5
GEM	Canadian Global Environmental Multiscale (model)
GPU	Graphics Processing Unit
HRES	High-Resolution (model)
HRRR	High-Resolution Rapid Refresh (model)
IFS	Integrated Forecasting System
NWP	Numerical Weather Prediction (model)
NAM	North American Mesoscale Forecast System (model)

1 *Guidance on Numerical Weather Prediction: Models and Data Sources.* EPRI, Palo Alto, CA: 2024: 3002027745.

and various multivariate regression methods to fit a pattern to a training dataset that contains explanatory inputs to the variable for which a prediction is needed. For example, in load forecasting, temperature, day of year, day of week, and time of day are typical inputs and load is the output. Similarly to AIWP, once the model is trained, it can be used to very quickly obtain future forecasts, but the technology is more mature, less complex, and requires far less computational power for initial training.

AIWP: EVOLUTION AND TRAINING

The initial development of AI weather forecast models was driven by large technology firms as part of a strategic drive to demonstrate the power of GPUs and AI in a realm that is well known to have computing power as a driving constraint. The models include GraphCast from Google DeepMind, Pangu-Weather from Huawei Cloud EI, FuXi from Fudan University, and FourCastNet from Nvidia. While each model is different in the details, they all utilize deep learning frameworks and train on a large amount of gridded weather data. The dataset typically used is the European Centre for Medium-Range Weather Forecasting (ECMWF) Reanalysis v5 (ERA5), which provides a four-dimensional *estimate* of global weather conditions for the period 1940 through the present day for the entire globe at an hourly time interval and roughly 31 km grid spacing, based on all available observations and short-range NWP modeling to fill the gaps.[2] Most AIWP models train with data from 1979 onwards, which is regarded as more accurate because it was after the advent of the satellite observing era.

In 2023, the developers of these models began to publicly make bold claims that their models were beating traditional NWP. The developers of GraphCast claimed they were beating the ECMWF high-resolution (HRES) model in many objective metrics. However, this is somewhat misleading because the comparisons used ERA5 as the validation "truth" and resampled the HRES output to ERA5 grid spacing. Furthermore, the AIWP models are trained to reproduce ERA5, while operational NWP models are tuned to produce the best forecast for their stakeholders. Nonetheless, the results were impressive, especially given that the HRES model is widely considered to produce the best global NWP forecasts and is a high-resolution operational implementation of the same Integrated Forecasting System (IFS) that is used in a lower resolution model to produce ERA5. Figure 1 provides a visual comparison of 2-meter (2-m) temperature from ERA5, HRES, and two AIWP models: GraphCast and Pangu-Weather.

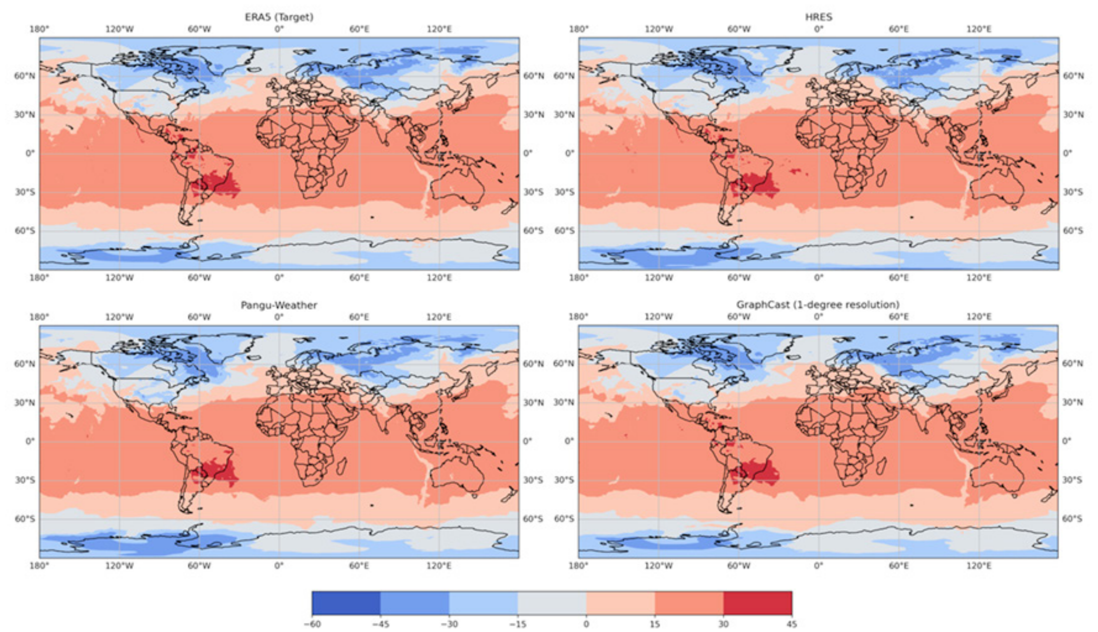

Figure 1. *Deterministic 6-hour ahead 2-m temperature forecast initialized at 2022-12-21 00:00:00 from HRES, Pangu-Weather and GraphCast compared against ERA5.*

2 The process of producing reanalysis data is complex and beyond the scope here. An introduction for non-meteorologists can be found in the ESIG Meteorology 101 companion to their Weather Inputs Report.

HOW IS AIWP CURRENTLY DEPLOYED AND WHAT IS THE STATE-OF-THE-ART?

Most AIWP systems are currently deployed in research mode, and while the available pre-trained AIWP forecast systems provide canned, off-the-shelf implementations, they cannot be customized. In addition, most of the models are currently global and relatively low resolution compared to operational NWP forecasts. The configuration (for example, variables predicted, resolution, and forecast interval) is mostly incorporated during the training process. Most changes require retraining, which is challenging, and requires powerful hardware and vast amounts of data. Meanwhile, an NWP model configuration can easily be modified according to the desired use case, for example, by using higher spatial and temporal resolution to capture finer scale weather features that might impact a wind farm or by employing a more sophisticated sub-model to handle aerosols that could impact a solar forecast. This process also requires considerable expertise and, for high-resolution simulations, substantial powerful hardware.

While training takes vast amounts of computational resource, a trained AIWP can produce a 10-day global forecast 1000–10,000 times faster than a conventional NWP model.[3] For example, such a forecast can be produced in about a minute on a single Tensor Processing Unit (TPU) v4, versus an NWP model taking several hours on a high-performance computing cluster. This opens the possibility of running large model ensembles, the output of which may revolutionize probabilistic forecasting and uncertainty quantification.

Despite their off-the-shelf nature, or perhaps because of it, the skill of AIWP models has advanced incredibly rapidly in the past two years. Most notably, in early 2024, the ECMWF announced that their new Artificial Intelligence Integrated Forecasting System (AIFS), which they began to develop in 2023, was already showing superior skill in many metrics than ECMWF's physics-based HRES model. Throughout 2024, scientists and the media lauded its performance, including its superior handling of tracking forecasts for hurricanes Helene and Milton. This is remarkable, because as already noted, HRES is widely regarded as the best global weather prediction system in the world by a significant margin (with objective measures to prove it).

Following are three other features of AIWP models worth mentioning:

- **Variables input and output:** AIWP produces forecasts from inferences between how weather fields in the training data evolve. Because they are data driven, some implementations do not have as many variables to be processed or information about as many levels in the atmosphere. For example, FourCastNet only produces data for a very limited set of core variables at the surface and on four pressure levels. This means that some models may not provide output variables that may be required for downstream use. However, this is not a fundamental limitation, and as models are operationalized, they are being updated to include the variables needed by stakeholders.
- **Smoothing:** Current AIFS global models are trained with ERA5 weather data, which have relatively coarse resolution. Because the training objective is the error in the forecast relative to the corresponding ERA5 time, output from the models does not contain the level of variability present in reality or in forecasts from equivalent NWP models such as HRES models.
- **Run-to-run continuity:** A major feature of AIWP models is that they make inferences at multiple scales and are less sensitive to errors in initial conditions. They will lock in on a solution and refine over subsequent runs. This is in contrast to the tendency for nonlinear error growth in NWP models to create divergent solutions, especially at long lead times. This feature was on dramatic display when the AIFS predicted Milton's landfall location to within 7 miles five days before landfall and stayed locked in within 13 miles, while NWP models experienced large swings.

The field continues to advance incredibly quickly, especially now that organizations such as ECMWF are melding the best meteorological knowledge with the most promising AI methods. For global forecasting, the trajectory is clear. The best AIWP models – trained with a combination of accurate global NWP model output such as the IFS with a spatial resolution of about 8 km – combined with reanalysis

3 Carissa Wong, "DeepMind AI accurately forecasts weather — on a desktop computer," *Nature,* 14 November 2023. DOI: https://doi.org/10.1038/d41586-023-03552-y

data covering many decades are superior to existing NWP on many metrics and will continue to improve. Research is also ongoing to tailor AIWP to regional and microscale applications that require weather impacts at smaller scales to be diagnosed. This work is in its infancy, but given the rate of evolution of global AIWP, it is reasonable to expect rapid advances in this area as well. Indeed, on December 24 ,2024, a paper was released describing the new OMG-HD model from Microsoft that provides regional AIWP. The model introduces a revolutionary way of assimilating weather data into the forecasting process, allowing more data to be ingested and avoiding some foundational limitations of NWP simulations. The paper was published on arXiv and thus has not been peer reviewed, so it is important that others validate and/or replicate the findings. Nonetheless, the results are compelling and indicative of a coming paradigm shift in weather prediction.

SUMMARY OF COMMON AI WEATHER FORECAST MODELS

Table 1 lists the specs of common global NWP models and AI weather models. A few notes regarding the listed models follow:

- The specifications for the first five global AIWP models listed here can be compared to those of operational global weather prediction models, including the HRES model from ECMWF, the Global Forecast System (GFS) from NOAA, the United Kingdom Met Office (UKMO) model, the German Weather Service Icosahedral Nonhydrostatic Model (DWD ICON), and the Canadian Global Environmental Multiscale (GEM) model.
- At the time of writing their resolution varies from about 9 km (HRES) to about 25 km (GEM). These models run out 10–15 days (15 days is about the limit of predictability due to the chaotic nature of the atmosphere). Output is hourly early in the forecast and reduced for longer horizons.
- Output interval is configurable in some of the AIWP models but in other cases requires complex processes and possible retraining.
- The number of variables predicted and the vertical levels on which they are output tends to provide less coverage on current AIWP models than NWP models. However, this is already changing as the major weather prediction centers begin implementing their AIWP models.
- Global NWP models take hours to run on massive high-performance computing (HPC) clusters. Pretrained AIWP models run in seconds to minutes.
- Regional models such as StormCast and OMG-HD can be compared with the likes of the North American Mesoscale (NAM) forecast system, and the High-Resolution Rapid Refresh (HRRR). The OMG-HD model is an AIWP that has a continuously refreshed assimilation cycle like the HRRR. Early results in this space indicate promise to produce high-resolution forecasts of comparable skill to NWP with much less computer power.

Table 1. *Major attributes of some common AIWP models.*

MODEL (SOURCE)	PRETRAINED MODEL WEIGHTS/ SOURCE CODE		PRETRAINED CHARACTERISTICS (EXTENT/ RESOLUTION/ INTERVAL)
AIFS	Yes: link	Yes: link	Global 0.25° (~28 km) 6 hours
Pangu-Weather (Huawei Cloud EI)	Yes: link	Yes: link	Global 0.25° (~28 km) 1/3/6/24 hours
GraphCast (Google DeepMind)	Yes: link	Yes: link	Global 0.25° (~28 km) 6 hours
FuXi (Fudan University)	Yes: link	No	Global 0.25° (~28 km) 6 hours
FourCastNet (Nvidia)	Yes: link	Yes: link	Global 0.25° (~28 km) 6 hours
ClimaX (Microsoft)	Yes: link	Yes: link	Global 1.40625° (~28 km) and 5.625° (~626 km) 6 hours
StormCast (Nvidia)	Not Determined		Continental United States (CONUS) 3 km 1 hour
OMG-HD (Microsoft)	No	No	CONUS 5 km 1 hour

IMPACT ON THE POWER SECTOR

Weather prediction is a very complex business that requires huge quantities of data to be gathered, managed, and manipulated, and this aspect will remain. However, most weather prediction meteorologists agree that AIWP is going to revolutionize forecasting, impacting all sectors where knowledge of future conditions is important, including the power sector. These developments will revolutionize the availability and use of weather information for power sector use in numerous other ways. In power system operations, the key impacts will be earlier delivery time, higher accuracy, more frequent updates, and better uncertainty quantification for all weather information and the products derived from it. More details of each of these aspects follow:

- **Speed:** AIWP front loads the heavy computing into the training component of the model process. Once trained, AIWP forecasts run orders of magnitude faster than NWP forecasts. The ECMWF's currently experimental AIFS will be operationalized in the second half of 2025. It is expected that this will result in forecasts that have many of the accuracy features of the HRES model but will be delivered in a fraction of the time after assimilation is complete (<30 minutes versus several hours). Initially, the big speed advance will be in global modeling, but output from global models feeds higher-resolution regional models. Results from models such as the OMG-HD suggest regional modeling will quickly change too. The forecast data from these models feed into many downstream processes and renewables forecasting, and the increased speed means more timely and current operational data (based on a recent versus older forecast).
- **Accuracy:** AIWP still needs to be carefully validated, and each methodology has its own nuance. However, initial results indicate that the method is less sensitive to initial condition error than NWP, and this, when combined with large training datasets, is yielding very accurate forecasts even though the method is in its infancy. New developments that can perform data assimilation directly from observations, thus skipping the NWP aspect of this process will likely further improve accuracy.
- **Uncertainty Quantification:** The speed of AIWP models allows massive ensembles to be performed. If well calibrated, these have the potential to allow the envelope of uncertainty to be estimated in ways that were not possible previously, including at longer lead times. This could have profound consequence for the power sector from enabling processes such as stochastic unit commitment to providing planning for the potential of high-impact events two weeks or more ahead.

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